

Finite-Catalog Renewal Design by Selected-Boundary Resource Paths

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Abstract. A renewal provider chooses a limited menu of terminal commands and allocates an accepted obligation across heterogeneous histories. We give an exact selected-boundary decomposition for finite catalogs with opening charges, individual expected caps, realization ceilings, and convex intermediate-service costs. Each adjacent selected pair owns a terminal-reward increment and the service costs of the histories whose eligibility ends there. The resulting resource path has enough capacity to restore the original obligation after rounding. With rational quadratic primitives, this identity yields a polynomial additive approximation scheme whose degree does not depend on the number of catalog-eligibility classes. The scheme retains the original command budget and admits independently checked rational global intervals. Exact polynomial common-prefix optimization remains available as a stronger special case. We establish catalog-insertion and threshold-stability results and extend the capacity-repair argument to compatible production modules. A fixed synthetic study tests strict interior prefixes through 1,024 histories, controlled eligibility splits, unused splitting commands, difficult historical models, certificate costs, and direct mixed-integer comparisons. Numerical optimization is faster on the small comparison models; the contribution is the structural reduction and its original-obligation certification guarantee.

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1 Introduction

A provider that serves heterogeneous histories may have to commit to a small common menu of terminal commands. The provider can adjust service before sending a command, but the terminal interface cannot recover the history from an uncharged side channel. Choosing the menu and allocating the accepted obligation are therefore coupled decisions. A command that improves terminal reward can require an opening payment, become unsafe in some histories, or displace another command from the interface. Optimizing the allocation after fixing the menu does not resolve this joint design problem.

We study finite-catalog renewal design with individual expected-obligation caps, history-specific realization ceilings, convex intermediate-service costs, and additive command-opening charges. The provider must meet the accepted obligation exactly and retain the original command budget. We establish a selected-boundary decomposition that turns the joint problem into an ordered resource-constrained path problem. A path chooses the commands. Each adjacent selected pair owns a terminal-reward increment and the service costs of precisely the histories whose eligibility ends at that pair. A terminal edge accounts for the remaining histories. Unselected catalog entries do not divide these ownership sets.

The decomposition yields an additive approximation scheme whose polynomial degree does not depend on the number of eligibility classes. A scalar resource grid replaces the earlier search over class means. Rounding an optimal path loses a controlled amount of resource; the selected path always has enough capacity to restore the exact obligation. This capacity identity is essential: a resource-grid solution alone would only solve a perturbed-obligation problem. We recover a feasible policy at the original obligation and certify its distance from the globally optimal finite-catalog value. The guarantee is additive, not relative, and does not enlarge the installed menu.

The result complements, rather than replaces, exact common-eligibility pooling. Histories that admit the same catalog prefix retain their individual caps and service costs under pooling. Their correction depends only on the last eligible selected command. With one eligibility class and rational quadratic primitives, a polynomial ordered-path calculation solves the joint problem exactly, including interior obligations and zero service curvatures. With arbitrary eligibility, the new resource path provides polynomial dependence on inverse normalized accuracy instead of a class-dependent exponent. We separate these two computational statements throughout.

Catalog dependence also becomes explicit. Inserting an unused command can split a full-catalog eligibility class without changing the optimal decision. By contrast, the resource contribution of an edge between two retained selected commands is unchanged by such insertion. We prove insertion and threshold-margin results, identify a discontinuity at a threshold crossing, and give robust value brackets for uncertain ceilings. These results distinguish genuine changes in the feasible interface from changes in a convenient partition.

Our analysis uses classical tools with specific roles. Separable convex allocation supplies service values and marginal-price witnesses; Patriksson (2008) surveys this problem and its algorithms. Resource discretization has a substantial history in constrained shortest-path approximation, including Hassin (1992) and Lorenz and Raz (2001). Totally monotone matrix search provides efficient convolution and independent Bellman verification (Aggarwal et al., 1987). We do not present these components as new. The contribution

is the exact selected-boundary reduction, the capacity identity that permits original-obligation recovery, and their consequence for finite-catalog joint design. The maximum over installed menus need not be concave, even though every fixed-menu response and every arc kernel is concave.

Finite policy graphs approximate continuation contracts by a finite recursive representation (Zhang, 2012). Our restriction instead concerns the shared terminal alphabet; the pre-draw allocation and the original equality remain explicit. Adjacent lotteries are related to mean-preserving random splitting in dual quantization (Pagès and Wilbertz, 2012), while ordered quantization can exploit matrix searching (Wu, 1991). Those connections motivate the representation, but do not by themselves resolve opening charges, history-specific safety prefixes, and endogenous allocation of the accepted obligation. The selected-boundary reduction addresses that joint decision.

The computational study is designed around the structural questions. It includes heterogeneous ceilings within strict interior prefixes, controlled eligibility splits, catalog-refinement paths, provably unused splitting commands, and every previously reported difficult instance. Exact enumeration supplies small-instance references. New resource-grid certificates are checked through a separate marginal-price construction and a different matrix-search algorithm. Matched-machine group and individual subdivisions, price-set sensitivity, and numerical mixed-integer envelopes expose the costs as well as the benefits of certification. The study reports certificate sizes, rational bit lengths, relative gaps, and unsuccessful historical requests. All inputs are synthetic; no claim of field calibration or general solver dominance is made.

The same capacity-repair argument applies to compatible production-module configurations with a divisible order and concave operating returns. We state this second model independently, without renewal histories or command lotteries. The archived continuous-design, institutional, and coarsening results remain available with their complete proofs and original assumptions. The present article concentrates on the finite-catalog joint design problem and its strengthened guarantees.

2 Finite-Catalog Renewal Design

There are k histories. History j has probability $\pi_j > 0$, expected-obligation cap $b_j \in [0, 1]$, and realization ceiling $\tau_j \geq b_j$, with $\sum_j \pi_j = 1$. Write $\bar{b} = \sum_j \pi_j b_j$. Intermediate service has cost $h_j : [0, b_j] \rightarrow \mathbb{R}_+$, convex and nondecreasing, with $h_j(0) = 0$. The common terminal reward $f : [0, 1] \rightarrow \mathbb{R}$ is strictly increasing and concave. The catalog is $\mathcal{A} = \{a_1 < \dots < a_N\} \subset [0, 1]$. Opening command a_i costs $\rho_i \geq 0$.

A nonempty selected book $c = (c_1 < \dots < c_s)$ has $s \leq m$. After observing j , the provider selects deterministic pre-draw service $y_j \geq 0$ and a lottery C_j supported on c . The problem is

$$V^*(B) = \max_{c, y, C} \left\{ \sum_j \pi_j [\mathbb{E}f(C_j) - h_j(y_j)] - \sum_{z \in c} \rho(z) \right\}, \quad (1)$$

$$t_j = y_j + \mathbb{E}C_j \leq b_j, \quad y_j + C_j \leq \tau_j \quad \text{almost surely}, \quad \sum_j \pi_j t_j = B. \quad (2)$$

Here $B \in [0, \bar{b}]$ is accepted, rather than optimized away. The common-risk specialization is $\tau_j = b_j + \delta$. Expected and realized obligations are different restrictions. The terminal symbol does not carry a free

history index, inherited service tier, or shared random seed. The cardinality m measures the command alphabet, not total implementation storage: targets, encoder logic, probability precision, program constants, and certificates are additional resources. The mathematical results below concern the stated finite catalog; continuous-design results have a separate approximation error.

For a feasible book, let $d_j = \max(c \cap [0, \tau_j])$, and let $F_{j,c}$ interpolate f linearly on this eligible prefix. A singleton prefix has the constant value $f(d_j)$ at its only point.

Proposition 2.1 (Canonical response). *A book is feasible if and only if $c_1 \leq \min_j b_j$ and $c_1 \leq B \leq \bar{b}$. At target $c_1 \leq t \leq b_j$, its optimal response is*

$$v_{j,c}(t) = F_{j,c}(\min\{t, d_j\}) - h_j((t - d_j)_+). \quad (3)$$

It is attained by $y_j = (t - d_j)_+$ and an adjacent eligible lottery with mean $\min(t, d_j)$. The response is concave in t .

Proof. Every feasible command satisfies $C_j \leq \tau_j$ since $y_j \geq 0$. At mean $\mu = \mathbb{E}C_j$, concavity on the ordered support bounds the reward by $F_{j,c}(\mu)$. The payoff is at most $F_{j,c}(\mu) - h_j(t - \mu)$ for $c_1 \leq \mu \leq \min(t, d_j)$. Both terms are nondecreasing in μ , so the upper endpoint maximizes this bound. When $t \leq d_j$, adjacent interpolation has zero intermediate service and support below τ_j . When $t > d_j$, use command d_j surely and service $t - d_j$; the realized sum is $t \leq b_j \leq \tau_j$. Thus the bound is attained without weakening either restriction. Interpolation slopes decrease and are nonnegative; the tail slopes of $-h_j$ are nonpositive and nonincreasing. This proves concavity. Feasible targets fill the entire weighted interval $[c_1, \bar{b}]$, proving necessity and sufficiency. $\square$

The algorithmic specialization is

$$f(x) = rx - \frac{1}{2}qx^2, \quad r \geq q \geq 0, \quad r > 0, \quad h_j(y) = \frac{1}{2}\gamma_j y^2, \quad \gamma_j \geq 0. \quad (4)$$

All probabilities, caps, ceilings, catalog points, shape parameters, opening charges, and requested accuracies are rationally encoded in exact-arithmetic claims. Optimization identities allow real nonnegative charges; bit-complexity assertions do not assign a finite encoding to arbitrary real numbers. A common Lipschitz constant for the canonical responses is

$$L = \max\{r, \max_j \gamma_j\} > 0. \quad (5)$$

A fixed-book allocation can be solved by filling positive chord slopes in decreasing order, then zero-cost service capacity, then quadratic service tails. The service breakpoints are $\gamma_j(b_j - d_j)_+$. This takes $O(k s + k \log(k + 1))$ rational operations with sorted marginal events. Alternatively, a direct supporting-price sweep verifies

$$\sum_j \pi_j v_{j,c}(t_j) = \lambda B + \sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}(t) - \lambda t\}. \quad (6)$$

Eligible knots, caps, and clipped stationary tail points suffice to check each scalar maximum. This certificate

proves optimality for that book, not over all books.

3 Pooling at Selected Boundaries

3.1 Terminal service precedes intermediate service

The first exchange argument does not require common catalog eligibility.

Lemma 3.1 (Global terminal-first allocation). *For a fixed book, an optimal canonical allocation cannot have both $t_i > d_i$ and $t_j < \min(b_j, d_j)$. Therefore terminal capacity is exhausted before positive intermediate service is assigned, even when histories have different eligible prefixes.*

Proof. Suppose both inequalities hold. Choose

$$0 < \delta \leq \min \left\{ t_i - d_i, \frac{\pi_j}{\pi_i} (\min(b_j, d_j) - t_j) \right\}.$$

Replace t_i by $t_i - \delta$ and t_j by $t_j + \pi_i \delta / \pi_j$. Weighted conservation is explicit: $-\pi_i \delta + \pi_j (\pi_i \delta / \pi_j) = 0$. The donor's terminal reward is unchanged and its intermediate cost cannot increase. The recipient remains in its terminal-only range and obtains a strictly larger interpolation reward. Both canonical policies satisfy their own ceilings by Proposition 2.1. The objective strictly increases, a contradiction. $\square$

For selected neighbors $u < v$, define the weighted terminal increment and the exiting histories by

$$T_{uv} = \sum_{j: \tau_j \geq v} \pi_j [\min(b_j, v) - u]_+, \quad (7)$$

$$J_{uv} = \{j : u \leq \tau_j < v\}, \quad c_j(u) = (b_j - u)_+, \quad Q_{uv} = \sum_{j \in J_{uv}} \pi_j c_j(u). \quad (8)$$

The interval is determined by the two *selected* endpoints, not by the full-catalog classes between them. Let

$$H_{uv}(y) = \min \left\{ \sum_{j \in J_{uv}} \pi_j h_j(z_j) : 0 \leq z_j \leq c_j(u), \sum_{j \in J_{uv}} \pi_j z_j = y \right\}, \quad 0 \leq y \leq Q_{uv}. \quad (9)$$

An empty set has capacity zero and cost zero at zero. Write $\sigma_{uv} = [f(v) - f(u)] / (v - u)$. The arc payoff is

$$\Phi_{uv}(x) = \sigma_{uv} \min(x, T_{uv}) - H_{uv}((x - T_{uv})_+), \quad 0 \leq x \leq T_{uv} + Q_{uv}. \quad (10)$$

For a terminal edge $(u, \dagger)$, put $T_{u\dagger} = 0$, $J_{u\dagger} = \{j : \tau_j \geq u\}$, and $\Phi_{u\dagger} = -H_{u\dagger}$, with the corresponding capacity $Q_{u\dagger}$. Every Φ is concave. Under (4), every arc is L -Lipschitz: terminal slopes are at most r , and service marginal costs are at most $\max_j \gamma_j c_j(u) \leq L$.

Theorem 3.2 (Exact selected-boundary resource path). *Fix a feasible book $c = (u_1 < \dots < u_s)$ with anchor*

$a = u_1$. Let $P(c)$ contain its $s - 1$ successive edges and its terminal edge. Then

$$\sum_{e \in P(c)} (T_e + Q_e) = \bar{b} - a, \quad (11)$$

and its optimized net value is exactly

$$V_c(B) = f(a) - \sum_{z \in c} \rho(z) + \max_{\substack{0 \leq x_e \leq T_e + Q_e \\ \sum_e x_e = B - a}} \sum_{e \in P(c)} \Phi_e(x_e). \quad (12)$$

Consequently, maximizing (12) over ordered paths with at most m selected vertices solves (1) exactly. The reduction introduces no target or eligibility-class grid.

Proof. Every history is eligible for the anchor because $a \leq b_j \leq \tau_j$. It exits at exactly one selected edge: the edge after its last eligible command, or the terminal edge. If its last eligible command is d_j , its terminal increments along earlier edges sum to $\min(b_j, d_j) - a$, and its unique exit-service capacity is $(b_j - d_j)_+$. Their sum is $b_j - a$. Multiplying by π_j and summing proves (11).

By Lemma 3.1, an optimal canonical allocation first uses terminal increments. The chord slopes of successive selected edges decrease by concavity of f , and are positive. It therefore fills the aggregate layer $T_{u_1 u_2}$, then $T_{u_2 u_3}$, and so on, with at most one partially filled layer. Filling an earlier layer never prevents a later one: every history eligible for a later command was eligible for each earlier selected command, and the individual increments telescope. Within the partial layer, distribute its mass among histories with the capacities in (7). If the obligation exceeds all terminal capacity, each history has terminal mean $\min(b_j, d_j)$ and its remaining service capacity belongs to its unique exit set. These sets are disjoint. Minimizing their total cost at the residual obligation is the infimal convolution of the functions H_e . This produces a feasible vector in (12) with the same value as the optimal original allocation.

Conversely, consider any feasible arc-resource vector. Interpret each x_e as terminal mass $\min(x_e, T_e)$ and service mass $(x_e - T_e)_+$. Redistribute mass to positive-slope terminal segments before service and to earlier selected chord layers before later ones. These exchanges weakly increase the sum of arc payoffs: terminal slopes are positive and decreasing, while service marginal returns are nonpositive. Once all terminal layers are full, service can be reallocated across the disjoint exit sets to minimize its cost. If terminal capacity is not exhausted, all service is removed and at most one terminal layer is partial. The preceding construction then implements this rearranged vector by original canonical policies with the same total obligation and at least its objective. Compactness ensures attainment. Both comparisons prove equality; the charges remain attached to exactly the selected commands. $\square$

Corollary 3.3 (Endogenous eligibility). *A book of s commands induces at most s nonempty eligible-prefix classes. Histories with different full-catalog prefixes but the same last eligible selected command can be pooled for that fixed book without replacing individual caps or costs. Adding unselected candidates leaves every arc function of a retained selected path unchanged.*

Proof. The nonempty exit sets in (8), including the terminal set, are exactly the selected-book classes and partition all histories. Their definitions and the terminal increments involve only selected endpoints and the

original primitives. □

The common terminal reward is substantive. History-dependent chord slopes need not share a common order; the rearrangement in Theorem 3.2 would then require additional structure. Likewise, a second independent aggregate equality is not eliminated by the scalar capacity identity. These assumptions specify the proved model rather than restrictions imposed to obtain a favorable experimental instance.

4 Joint Optimization with One Resource Grid

4.1 A capacity-repair principle

We first state the approximation argument for a general resource-path problem. It identifies exactly which property is supplied by Theorem 3.2.

Consider a finite acyclic directed graph with designated source and sink. An admissible path has at most h arcs. Arc e has capacity $C_e \geq 0$, opening charge κ_e , and an L -Lipschitz payoff ψ_e on $[0, C_e]$. Choose a path and resources satisfying $\sum_e x_e = R$, to maximize $\sum_e [\psi_e(x_e) - \kappa_e]$. Assume every admissible path has total capacity at least $R \geq 0$. This assumption is verifiable by a minimum-capacity path calculation with the same length restriction; it is automatic when $C_{uv} = p_v - p_u$ for a nondecreasing vertex potential and $R \leq p_{\text{sink}} - p_{\text{source}}$.

For $\eta > 0$, restrict resources to $x_e = q_e \eta$, where q_e is a nonnegative integer and $q_e \eta \leq C_e$. Accept a grid path only if

$$0 \leq R - \eta \sum_e q_e \leq h\eta. \quad (13)$$

Let S_η be the largest grid-path score in this band.

Theorem 4.1 (Capacity-repair certificate). *A maximizing grid path can be repaired to exact resource R without changing its arcs. If its repaired feasible value is $\underline{V}$, the optimum V satisfies*

$$S_\eta - Lh\eta \leq \underline{V} \leq V \leq S_\eta + Lh\eta. \quad (14)$$

Consequently the returned path is within $2Lh\eta$ of optimal. The result permits signed payoffs and arbitrary fixed charges; it does not use a positive lower bound on the optimal value.

Proof. Round each resource on an optimal path down to a multiple of η . The lost resource d is less than $h\eta$, so the rounded vector satisfies (13). Lipschitz continuity bounds its objective loss by Ld , because charges do not change. Hence $V \leq S_\eta + Lh\eta$.

For the maximizing grid path, its unused total capacity is at least $R - \eta \sum_e q_e$. Distribute exactly that residual among its nonnegative remaining capacities. No coordinate decreases and the sum of absolute resource changes equals the residual, which is at most $h\eta$. The payoff loss is at most $Lh\eta$. The repaired vector is feasible for the original resource requirement and uses the same path. This proves all inequalities. □

Concavity is not needed for this error argument. It will give efficient convolution and reconstruction in the renewal problem. An arbitrary resource-constrained path problem with some accepted grid paths of

insufficient total capacity would not satisfy this theorem; its rounded solutions could be infeasible to repair. The identity (11) rules out precisely that failure here.

4.2 Renewal recurrence and exact-obligation recovery

For each feasible catalog anchor $a_h \leq \min(B, \min_j b_j)$, set $R_h = B - a_h$ and $Q_h = \lfloor R_h / \eta \rfloor$. Let $D_{\ell,v}^h(z)$ be the best score of a path that starts at a_h , ends at a_v , has ℓ selected commands, and consumes resource $z\eta$ on its nonterminal edges. The initial condition is

$$D_{1,h}^h(0) = f(a_h) - \rho_h.$$

Unreachable states have value $-\infty$. For $\ell \geq 2$,

$$D_{\ell,v}^h(z) = \max_{\substack{u < v, 0 \leq t \leq z \\ (z-t)\eta \leq T_{uv} + Q_{uv}}} \{D_{\ell-1,u}^h(t) + \Phi_{uv}((z-t)\eta) - \rho_v\}. \quad (15)$$

Complete each state with its terminal edge. Retain totals z in the band

$$\max \left\{ 0, \left\lceil \frac{R_h - m\eta}{\eta} \right\rceil \right\} \leq z \leq Q_h. \quad (16)$$

The maximum over all anchors, all $1 \leq \ell \leq m$, and all terminal completions is S_η . The terminal edge counts among the at most m rounded edges. It does not consume another command slot or incur another command charge.

Theorem 4.2 (Polynomial additive finite-catalog design). *Under (4), let the inputs and charges be rational, and choose $\eta = \varepsilon / (2Lm)$ for $\varepsilon > 0$. Recover an optimal fixed-book policy at the original promise B from a maximizing grid path. Its value $\underline{V}$ and the bound $\bar{V} = S_\eta + Lm\eta$ satisfy*

$$\underline{V} \leq V^*(B) \leq \bar{V}, \quad 0 \leq \bar{V} - \underline{V} \leq \varepsilon. \quad (17)$$

All original individual caps, realization ceilings, opening charges, and the command budget are respected. With $Q = \lceil 2Lm/\varepsilon \rceil + 1$, a direct exact implementation uses

$$O(kN^2Q \log(k+1) + mN^3Q \log(Q+1)) \quad (18)$$

arithmetic operations. Its bit complexity is polynomial in k, N, m, Q and the binary input length. Neither count has an exponent depending on the number of eligibility classes.

Proof. For every feasible anchor, Theorem 3.2 gives capacity $\bar{b} - a_h \geq R_h$ on every path. Its arc functions have Lipschitz constant L . Apply Theorem 4.1 with at most m edges, including the terminal edge. A repaired resource solution can be implemented with no smaller payoff by the rearrangement in Theorem 3.2. Exact fixed-book optimization at B can only improve this feasible reconstruction. Every optimal original book has an enumerated feasible anchor, so maximizing across anchors gives (17).

There are $O(N^2)$ possible edges, including terminal edges. Each has at most Q grid values. Solving its capped quadratic allocation by sorting at most k marginal breakpoints costs $O(k \log(k + 1))$ operations per value; this conservative preprocessing bound matches the implemented reference construction. The monotone convolution below evaluates one edge transition in $O(Q \log(Q + 1))$ operations. There are at most N anchors and mN^2 transitions per anchor. Terminal completions are included in the same bound.

Every grid value is rational with numerator and denominator length polynomial in the input length and k and $\log Q$. The least common multiple of the finitely many grid denominators, anchor rewards, and charges has bit length no greater than the sum of those lengths, hence polynomial in k, N, Q and input length. Scaling by this *exact* denominator converts all Bellman additions and comparisons to integer arithmetic without payoff quantization. Summing at most m terms preserves polynomial bit length. Recovery and the final bound use rational arithmetic as well. $\square$

For normalized accuracy $\epsilon = \varepsilon/L$, $Q = O(m/\epsilon)$. Thus the theorem is polynomial in inverse normalized accuracy even when L is encoded in binary. For a prescribed absolute error, the dependence is on the numerical ratio L/ε , not on its logarithm. This is an additive scheme, not a relative-error fully polynomial approximation scheme. Large opening charges or an optimum near zero do not invalidate the certificate.

4.3 Concave kernels, not concave Bellman values

The sequence $K_e(q) = \Phi_e(q\eta)$ is concave. For an arbitrary previous-state vector $A(t)$, the transition matrix

$$M(z, t) = A(t) + K_e(z - t)$$

is anti-Monge on its feasible band: the column terms cancel, and decreasing successive differences of K_e give the quadrangle inequality. Its leftmost row maximizers are nondecreasing. A divide-and-conquer row search therefore evaluates the max-plus convolution in $O(Q \log(Q + 1))$ comparisons. The previous vector need not be concave. All reachable resource indices form a prefix, because each edge admits every integer from zero to its capacity index; taking the maximum over paths preserves this contiguous support.

This distinction is necessary. With one history, cap one, zero service cost, $f(x) = x$, catalog $\{0, 1/2\}$, budget two, and charges $\rho(0) = 0$, $\rho(1/2) = 1/10$, the optimized value is

$$V^*(B) = \max\{0, \min(B, 1/2) - 1/10\}, \quad 0 \leq B \leq 1.$$

Its slope increases at $B = 1/10$, so it is not concave. A single supporting price need not certify the global menu choice. The resource grid avoids that incorrect concavity inference.

4.4 Independent global certificates

A resource certificate contains the original model, grid step, common denominator, all feasible-anchor Bellman tables, the best band score, and an original-space lottery policy. The checker imports neither the new optimizer nor an antecedent optimizer. It reconstructs service values by enumerating marginal-price

intervals. For an interval with saturated cost c_0 , saturated mass y_0 including free service, and active reciprocal curvature $R_0 = \sum_{j \in A} \pi_j / \gamma_j$, it uses

$$\alpha = (y - y_0) / R_0, \quad H(y) = c_0 + \frac{1}{2} R_0 \alpha^2,$$

checks that α lies in the stated interval, and evaluates every arc grid value exactly. Free service and zero capacities are handled separately.

It then checks every Bellman row using totally monotone matrix search (Aggarwal et al., 1987), rather than the optimizer's divide-and-conquer recurrence evaluator. For a finite kernel and previous vector, a sufficiently steep concave linear continuation outside the kernel support makes all infeasible entries strictly inferior to a feasible one. This permits ordinary rectangular matrix search without treating equal negative-infinity entries as valid witnesses. The checker verifies the complete anchor and state sets, the terminal band maximum, and $\bar{V} = S_\eta + Lm\eta$. Finally, it checks nonnegative lottery probabilities, unit mass, all realized ceilings, individual expected caps, exact root equality, the selected-book charge, and the stated interval width. A completion flag is never used as evidence.

Storage is $O(mN^2Q + N^2Q)$ rational or scaled-integer entries when all anchor tables are retained; an optimizer that does not retain certificates can process anchors sequentially. Verification cannot be sublinear in the supplied certificate size. The experiments report both storage and checking costs rather than counting only the command alphabet.

5 Exact Common Eligibility and Catalog Stability

5.1 The exact one-class result remains available

For the full catalog, define $K_j = |\mathcal{A} \cap [0, \tau_j]|$. Histories with equal K_j form a catalog-eligibility class G_g ; let E be their number, $W_g = \sum_{j \in G_g} \pi_j$, and $w_{j|g} = \pi_j / W_g$. These classes are sufficient for a class-mean algorithm, not an intrinsic dimension of every optimal selected book.

Fix one such class and a common eligible selected prefix with last level d . At mean s , define capped equalization $\theta_j(s) = \min(b_j, z)$, where $\sum_{j \in G_g} w_{j|g} \min(b_j, z) = s$. Let

$$D_g(d) = \sum_{j \in G_g} w_{j|g} \min(b_j, d), \quad C_g(y; d) = \min_{\substack{0 \leq z_j \leq (b_j - d)_+ \\ \sum_j w_{j|g} z_j = y}} \sum_{j \in G_g} w_{j|g} h_j(z_j).$$

The common terminal interpolant is $F_{g,c}$.

Proposition 5.1 (Exact class pooling). *For $c_1 \leq s \leq \sum_j w_{j|g} b_j$, the optimized class value is*

$$V_{g,c}(s) = \begin{cases} \sum_j w_{j|g} F_{g,c}(\theta_j(s)), & s \leq D_g(d), \\ \sum_j w_{j|g} F_{g,c}(\min(b_j, d)) - C_g(s - D_g(d); d), & s \geq D_g(d). \end{cases} \quad (19)$$

Both formulas agree at $D_g(d)$ and the value is continuous there. Relative to evaluating the canonical

response at $\theta(s)$, the exact correction is

$$\Delta_g(d; s) = \sum_{j \in G_g} \pi_j h_j((\theta_j(s) - d)_+) - W_g C_g((s - D_g(d))_+; d) \geq 0. \quad (20)$$

It depends on the book only through its last eligible selected level.

Proof. Below $D_g(d)$, Lemma 3.1 rules out intermediate service at an optimum. Capped equalization maximizes the sum of the same concave interpolant: among uncapped coordinates, equal levels satisfy the supporting-slope inequalities; capped smaller coordinates have no feasible upward movement, and the same common supporting slope supports their upper-bound constraint. This proves the first formula, also when the interpolant has affine pieces and the maximizing allocation is not unique.

Above $D_g(d)$, all terminal capacities must be filled and the residual is the stated capped-cost problem. At the junction, its resource and cost are zero, and $\theta_j(D_g(d)) = \min(b_j, d)$, so both expressions coincide. Continuity follows from continuity of interpolation and the finite capped convex value function. If $s > D_g(d)$, the equalization water level exceeds d ; otherwise its weighted mean would be at most $D_g(d)$. Thus $\theta_j(s) \geq \min(b_j, d)$ coordinatewise, and

$$\sum_j w_{j|g} (\theta_j(s) - d)_+ = s - D_g(d).$$

The equalized service vector is feasible for C_g , which proves (20). For $s \leq D_g(d)$ both service terms vanish. $\square$

For completeness, the fixed-target path to which this correction is applied is explicit. For a target vector t , put

$$A_t(u, v) = \sum_{j: u \leq t_j < v} \pi_j \begin{cases} f(u) + (t_j - u)\sigma_{uv}, & v \leq \tau_j, \\ f(u) - h_j(t_j - u), & v > \tau_j, \end{cases}$$

$$Z_t(u) = \sum_{j: t_j \geq u} \pi_j \{f(u) - h_j(t_j - u)\}.$$

Every target is owned by its unique bracketing selected edge, or by the terminal node. Hence its book value is the sum of these terms less the selected charges. A backward longest-path recurrence with a command counter optimizes the book in $O((k + m)N^2)$ operations.

Theorem 5.2 (Exact common-prefix joint design). *If $E = 1$, one fixed-target path calculation at $t_j = \theta_j(B)$, corrected by (20), solves (1) exactly. In the rational quadratic case its arithmetic cost is*

$$O((k + m)N^2 + kN \log(k + 1)).$$

It permits heterogeneous caps, heterogeneous costs including zero curvatures, rational nonnegative opening charges, and any feasible promise.

Proof. The sole class mean is fixed at B . For each possible last eligible level, Proposition 5.1 gives the exact difference between equalized-target evaluation and optimal class allocation. Along a selected path, add this correction once: to the edge that leaves the eligible catalog prefix, or to the terminal node if no selected edge leaves it. Every book is represented and corrected exactly once, without changing its charge or cardinality. Maximizing the corrected path values therefore maximizes the true joint objective. Compute the N candidate capped-cost corrections by rational water-level sweeps in $O(kN \log(k+1))$ operations. The fixed-target path supplies the remaining bound. All input denominators and sorted rational events have polynomial encoding length. $\square$

For several classes the same construction is exact at *supplied* class means. The resource-path scheme avoids searching these means and therefore strengthens, rather than weakens, the variable-eligibility result. Exact enumeration remains polynomial for each fixed command budget but has an exponent depending on that budget. The new additive scheme is polynomial in the budget itself and inverse normalized accuracy. No claim of an exact unrestricted polynomial algorithm, a parameterized hardness lower bound, or a relative-error approximation follows from these statements.

5.2 Insertion and threshold perturbation

Proposition 5.3 (Catalog and selected-book stability). *Insert one candidate into a catalog, leaving every original point and charge unchanged. The full-catalog class count changes from E to either E or $E+1$; after r insertions it is at most $\min(k, E+r)$. For every retained selected book, its resource arcs and fixed-book value are unchanged. If no ceiling perturbation crosses a catalog candidate and each perturbed ceiling remains at least its cap, all feasible canonical responses and the global value are unchanged.*

Proof. Only the histories in the old catalog gap containing the inserted point can split an old class. They form at most one old class. Other classes retain their members and possibly acquire a new prefix-length integer. Thus there is at most one additional class per insertion. Fixed-book arc invariance is Corollary 3.3. If no ceiling crosses any candidate, each history has the same eligible prefix for every book; Proposition 2.1 then gives identical feasible targets and payoffs. $\square$

An insertion can therefore increase E without changing the optimum. Given an incumbent value v_0 , any new command with $\rho(z) > f(1) - v_0$ is dominated: a book using it has net value at most $f(1) - \rho(z) < v_0$, because all service costs and other charges are nonnegative. A dominated point inserted between two ceilings can split their full-catalog class, yet it cannot appear in an optimal book. The resource functions of every retained optimal path stay identical. A previously certified global upper bound must nevertheless be recomputed over the enlarged path set unless this dominance argument certifies exclusion of every added option.

Threshold margins matter. If $\min_{j,i} |\tau_j - a_i| > \delta$ and perturbations have size at most δ , with strict noncrossing at endpoints, the global value is invariant. Without a margin there need not be a vanishing error bound. For one history with $b = B = 1/4$, catalog $\{0, 1/2\}$, budget two, zero charges, $f(x) = 2x - x^2/2$, and $h(y) = y^2/2$, a ceiling just below $1/2$ gives value $-1/32$. At ceiling $1/2$, a half-and-half lottery gives $7/16$. The jump is $15/32$, although the ceiling change can be arbitrarily small.

For uncertain ceilings $\tau^- \leq \tau \leq \tau^+$, with $\tau^- \geq b$, monotonicity of the original feasible set gives

$$\underline{V}(\tau^-) \leq V^*(\tau) \leq \bar{V}(\tau^+). \quad (21)$$

The lower policy is feasible throughout the uncertainty set. Thus endpoint certificates give robust decisions without incorrectly assuming continuity across a command threshold. Post-optimization pooling may compress a fixed-book allocation certificate, but it must retain the original individual cap and service witnesses; it cannot certify omitted alternative books by itself.

6 A Second Model: Compatible Production Modules

The capacity-repair theorem is not specific to renewal lotteries. Consider a production interface assembled from compatible modules. A directed acyclic graph describes compatibility; choosing a source-to-sink path installs its modules, with at most h modules and installation payment $\kappa_e \geq 0$ for module e . Module e can process a divisible amount $x_e \in [0, C_e]$ and earns concave net operating return $\psi_e(x_e)$. An accepted order of size R must be assigned exactly:

$$\max_{P, x} \left\{ \sum_{e \in P} [\psi_e(x_e) - \kappa_e] : |P| \leq h, \sum_{e \in P} x_e = R, 0 \leq x_e \leq C_e \right\}. \quad (22)$$

Modules on a selected compatibility chain supply parallel divisible capacity; the order is allocated across them, not passed serially through each module. This distinguishes the workload equality in (22) from a serial-flow bottleneck model.

One relevant design is capacity-conservative configuration. Each graph vertex has a cumulative capacity coordinate p_v , and a module spanning u to v supplies $C_{uv} = p_v - p_u \geq 0$. A complete chain covers the same total capacity regardless of how finely it is configured. The operating curves and installation charges may differ across compatible modules. For example, $\psi_e(x) = \alpha_e x - \beta_e x^2/2$, with $\beta_e \geq 0$, represents diminishing operating return; it need not be increasing over the entire capacity range.

Corollary 6.1 (Original-order production guarantee). *Suppose every admissible configuration in (22) has total capacity at least R , and all operating curves are L -Lipschitz. Resource gridding with step $\eta = \varepsilon/(2Lh)$ and residual repair returns a feasible configuration and an additive interval of width at most ε , without exceeding the module budget. If the curves are concave and rational grid values can be evaluated in polynomial time, the length-indexed dynamic program has $O(h|\mathcal{E}|Q \log(Q+1))$ arithmetic complexity after preprocessing, where $Q = \lceil R/\eta \rceil + 1$.*

Proof. Every accepted rounded configuration can absorb its omitted workload by the capacity assumption. Theorem 4.1 therefore applies directly. A topological dynamic program uses the same scalar-resource convolution as (15), now over graph arcs rather than catalog pairs and without anchor enumeration. Concavity gives the monotone convolution bound. Capacity-conservative configurations satisfy the assumption by telescoping $\sum_{e \in P} C_e = p_{\text{sink}} - p_{\text{source}}$. $\square$

This second model has neither heterogeneous renewal histories nor terminal-command lotteries. It reuses the capacity-repair mechanism, whereas Theorem 3.2 derives that mechanism from the original renewal constraints. We offer the model as a mathematical application of the abstraction, not as an observed industrial deployment. Empirical calibration of either setting requires operational data beyond the synthetic study reported here.

7 Computational Study

7.1 Design and independent validation

The study separates exact structural solvability, additive certification, and numerical optimization. Its protocol fixes all inputs, seeds, tolerances, and comparisons before the full run. It discloses the small development pilots rather than treating them as unseen tests. The original six difficult models are retained at two requested accuracies. Twelve new strict-prefix cases vary history count, catalog size, budget, and promise; six cases vary catalog eligibility under otherwise matched primitives; fourteen cases test within-gap perturbations, threshold crossings, free refinement, and dominated splitting commands. All 44 requests are reported.

Every resource-path interval is checked independently in original policy space. Exact enumeration over books, using a separate antecedent fixed-book allocator, supplies references for all 32 resource-grid cases. The twelve large common-prefix cases instead use the exact pooling theorem and a separate primal-dual and complete-path checker; exhaustive enumeration is not claimed at that scale. A further 36 group-subdivision certificates are checked. This gives 80 independently checked certificates across the principal study and its baseline comparisons.

Regression tests compare the selected-boundary identity with original fixed-book optimization on 1,583 books from 100 seeded models. All 100 corresponding global intervals contain their exhaustive optima. Two thousand concave-kernel convolution cases compare brute force, the optimizer’s divide-and-conquer search, and the checker’s totally monotone search. Twenty combinatorial tests exercise the dyadic count, including the ratio-three example. Twelve deliberately corrupted certificates are rejected. These finite tests supplement the proofs; they do not substitute for them.

All inputs are synthetic. The computational record includes the exact model vectors, original policies, solver statuses, complete certificate tables, environment, and content hashes. Timing records distinguish optimization from checking. Optimization includes payoff preprocessing, path search, recovery, and table construction; serialization and compression are excluded and disclosed separately. Publication reruns retain the same protocol and generate the displayed tables directly from their execution records. Local development measurements remain separately labeled.

7.2 Difficult models and competitive comparisons

Table 1 shows the six models at tolerance 0.001. Every resource-path interval meets the request; the largest width is 0.000500. Optimization takes 3.523–19.762 seconds in the recorded environment. Every returned

lower policy equals the exhaustive reference. This latter fact does not turn a positive-width resource certificate into a zero-width global proof: exact attainment is known here from the separate exhaustive comparison.

Absolute accuracy can mask material relative uncertainty. The largest reported relative width is 1.63%, on a small positive reference value. We therefore report the reference, absolute width, and relative width together, and retain width divided by the primitive scale in the machine-readable record. No economic importance is assigned to a numerical tolerance without application-specific calibration.

Table 2 compares class-mean subdivision at node allowances 1, 7, 31, and 127, root-price refinement at three settings, and individual-box subdivision. At 127 nodes with the default price setting, 2 of six requests attain 0.001. The original historical record had ten unresolved fine-tolerance requests among its two node budgets; every such request remains archived unchanged. Increasing the root-price set helps some cases but can alter the adaptive cover and worsen another finite-budget interval. We report that outcome rather than selecting the best setting after observing the answer.

These subdivision comparisons use the same machine but fixed node allowances, not matched time and memory ceilings. The direct mixed-integer comparison below shares a measured optimization-time allowance with the resource method. We do not claim a complete equal-memory ranking, and we have not added node-local price generation or an exhaustive repaired-net timing study. The stronger variable-eligibility algorithm removes the need to rely on those heuristics for the new polynomial guarantee.

7.3 Nontrivial eligibility and catalog refinement

Table 3 reports exact common-eligibility instances with 16 through 1,024 histories. All ceilings lie strictly inside the same interior catalog gap, differ across histories, and exclude a nonempty suffix. Thus the result is not restricted to the all-ceilings-equal-one geometry. Each case retains nonuniform weights, heterogeneous caps and opening charges, and zero as well as positive quadratic curvatures. Promises include interior and upper-bound obligations. All twelve cases close with zero rational width at the root.

Table 4 varies the full-catalog class count from one to twelve while holding the other primitives fixed. The generic resource method is run at each count, including the one-class case, to isolate its behavior. The grid dimension does not grow with that class count; edge capacities and exact arithmetic still change, so identical runtime is neither expected nor asserted. The two-class returned policy is below its exact reference by 83/143360; the other five attain their references. This distinguishes the additive guarantee from exact attainment. The study remains modest in catalog size and does not establish large-instance superiority for variable eligibility.

The refinement experiment inserts four commands into a four-point catalog. With opening charge three, each insertion is provably dominated, yet the full-catalog class count rises from one to five. The winning two-command book, its single selected eligibility class, and its exact value remain unchanged. With free insertions, the first command improves the value and the next three split classes without further improvement; the winning book has only two selected classes. Within-gap ceiling perturbations preserve the value exactly. The near-boundary pair changes full-catalog eligibility but retains its winning book; the analytical example in Section 5 separately demonstrates that a true value jump is possible. The companion reports all fourteen cases, not only those with a changed optimum.

7.4 Certificate cost and mixed-integer diagnostics

Table 5 reports certificate economics on the fine-tolerance models. Compressed sizes range from 0.216 to 6.628 decimal megabytes; the largest recorded denominator has 553 bits. The checker reconstructs the arc values and checks all anchor tables, so certification carries a real storage and computation cost. A small terminal alphabet is not a bound on the target table, encoder logic, probability precision, or audit storage.

Table 6 exposes the numerical envelope results rather than summarizing only their completion status. The implementation uses SciPy’s mixed-integer interface (Virtanen et al., 2020) on a direct support-indicator probability/service formulation, independently of the resource-path reduction. All twelve envelope solves finish successfully within their assigned allowances. They are much faster than the fine resource-grid construction on these small instances; the largest paired runtime is 0.038 seconds. Exhaustive fixed-book enumeration is also competitive at this scale. The case for the resource method is its proved additive complexity and exact original-space certificates, not a universal speed ranking.

The numerical upper and lower values agree to floating-point precision on these cases, including one tiny negative raw width from rounding. We preserve that diagnostic rather than relabeling it an exact proof. Envelope approximation allowances, solver relative gaps, model residuals, branch-and-bound nodes, and both objectives remain in the record. None of these floating-point statuses is used by the rational checker. A larger calibrated operational comparison would be needed to decide when the exact certificate cost is justified in practice.

8 Conclusion

Selecting a finite command menu and allocating an accepted obligation can be optimized through the boundaries of the selected menu itself. Terminal-first allocation separates positive terminal layers from intermediate service. Each selected edge owns the histories whose eligibility ends there, and every path has exactly the capacity required for original-obligation repair. This identity yields a polynomial additive scheme for arbitrary eligibility while retaining exact common-prefix optimization as a stronger special case.

The guarantee preserves the original obligation, caps, realization ceilings, charges, and command budget. Its dependence on inverse normalized accuracy is polynomial; it is not an exact unrestricted algorithm or a relative-error approximation. A second capacity-conservative production model identifies the part of the argument that extends beyond renewal contracts. Catalog and threshold analysis distinguishes partition changes from genuine feasible-set changes.

The theory and the synthetic evidence answer different questions. Exact certificates establish feasible decisions and valid global intervals. They do not establish that a small command interface is the principal implementation cost, that thresholds are accurately estimated, or that the method is the fastest optimizer. The retained numerical comparisons and explicit certificate costs make those distinctions operationally visible.

Code and Data Availability

The accompanying source package contains the fixed protocol, exact input instances, both optimizers and independent checkers, all execution records, and instructions for reproducing every table. The main implementation uses standard-library rational arithmetic; the mixed-integer comparison additionally uses SciPy. An immutable archive retains every antecedent manuscript, proof, and result under its original assumptions, together with a content-location map. These historical files are not prerequisites for reading the present proofs and are not silently replaced by the new article. The repository README identifies the current revision, parent commits, and the code-and-data archive. No empirical data or confidential application records were used.

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Table 1: Difficult instances: exact-resource intervals at tolerance 0.001

Seed	E	$E(c)$	Reference	Width	Relative	Seconds
49500	6	2	0.302083	0.000500	0.17%	13.831
49501	7	2	0.535640	0.000470	0.09%	18.391
49502	6	2	0.459188	0.000490	0.11%	17.073
49503	8	2	0.030152	0.000493	1.63%	3.523
49504	5	1	0.464255	0.000420	0.09%	15.290
49505	6	2	0.436758	0.000430	0.10%	19.762

All instances have 12 histories, 9 candidates, and a command budget of 3. E counts full-catalog classes and $E(c)$ counts classes on the returned book. Relative width is the interval width divided by the positive exhaustive reference. Every returned lower policy attains the exhaustive value, but its resource-path certificate has positive width.

Table 2: Matched-machine subdivision and price sensitivity

Method	Nodes	Steps	Complete	Median width	Largest width	Seconds
Group	1	4	0	0.003165	0.026387	0.095
Group	7	4	0	0.003165	0.011484	0.321
Group	31	4	0	0.002099	0.009043	1.225
Group	127	4	2	0.001398	0.005906	4.438
Group	31	0	0	0.116023	0.170215	0.260
Group	31	8	3	0.001242	0.004110	0.671
Individual	31	4	0	0.003165	0.010020	0.598

Each row contains all six seeds in Table 1. Complete means an interval width at most 0.001. Nodes are the allowed node count, not a common time or memory allowance. Steps is the inherited root-price refinement parameter; zero retains only zero price. Seconds is the median optimization time. With a finite adaptive cover, a changed price set may change the chosen subdivisions, so more steps need not dominate every earlier interval.

Table 3: Exact common eligibility with a strict interior prefix

Histories	Design 1	Design 2	Design 3	Check/solve	Width
16	0.018	0.067	0.074	1.530	0
64	0.060	0.236	0.258	1.619	0
256	0.230	0.910	0.985	1.677	0
1024	0.916	3.598	3.914	1.686	0

Optimization times are in seconds. Designs 1–3 use, respectively, $(N, m, B/\bar{b}) = (13, 2, 0.5), (25, 3, 0.9), (25, 5, 1)$. Every ceiling differs across histories and lies strictly in the catalog gap immediately above $1/2$; a nonempty catalog suffix is ineligible. Caps, weights, costs, and charges are heterogeneous, including zero curvature. The ratio is the largest checker-to-solver ratio across the three designs. These are structural zero-width closures, not extrapolated results for arbitrary eligibility.

Table 4: Controlled full-catalog eligibility sweep

E	$E(c)$	Width	Width/ L	States	Seconds
1	1	0.004449	0.002225	18,671	2.010
2	2	0.004560	0.002280	18,671	2.783
3	2	0.004449	0.002225	18,671	2.986
4	3	0.004661	0.002331	18,671	3.130
6	3	0.005006	0.002503	18,671	3.227
12	3	0.005052	0.002526	18,671	3.411

All six instances share 12 histories, 17 candidates, budget 3, promise fraction 0.85, caps, weights, costs, and charges. Only the ceiling grouping changes. The tolerance is 0.01. The generic resource method is used even at $E = 1$ to isolate its scaling; the exact common-prefix algorithm remains available. Every interval contains its independent exhaustive reference. At two classes the returned lower value is below that reference by 83/143360; the other five returned values equal their references.

Table 5: Resource certificate economics on the difficult instances

Seed	Packed	Plain	Entries	Bits	Check/solve
49500	1.737	4.150	75,647	174/174	0.162
49501	2.360	5.526	94,591	185/185	0.156
49502	4.041	9.041	89,731	327/327	0.154
49503	0.216	0.595	17,546	104/104	0.171
49504	5.555	12.175	80,444	495/495	0.148
49505	6.628	14.469	85,753	553/553	0.146

Packed and Plain are decimal megabytes of compressed and uncompressed JSON. Entries counts serialized rational or common-denominator integer values. Bits reports the maximum numerator length and maximum denominator length; Bellman numerators are measured before reducing their common denominator. Verification reconstructs all arc kernels and checks every anchor table. Storage and checking are part of the certification cost, not part of the command alphabet budget.

Table 6: Direct mixed-integer envelope diagnostics

Seed	Tangent value	Upper	Secant value	Allowance	Seconds	Width
49500	0.302083	0.302083	0.302083	$1.12 \cdot 10^{-4}$	0.038	0
49501	0.535640	0.535640	0.535640	$1.32 \cdot 10^{-4}$	0.031	0
49502	0.459188	0.459188	0.459187	$4.32 \cdot 10^{-5}$	0.015	$5.55 \cdot 10^{-17}$
49503	0.030152	0.030152	0.030152	$5.91 \cdot 10^{-5}$	0.034	$-1.39 \cdot 10^{-17}$
49504	0.464255	0.464255	0.464255	$2.77 \cdot 10^{-5}$	0.015	0
49505	0.436758	0.436758	0.436758	$8.83 \cdot 10^{-5}$	0.020	0

Each formulation has 249 variables, including 117 binaries, and 746 constraints, with 32 service-envelope segments. All 12 solves return status zero and one branch-and-bound node. Their recorded relative solver gaps are below 10^{-12} . The two solves share an allowance equal to the resource solver time, subject to a minimum of 0.1 seconds per solve. Allowance is the analytic envelope approximation allowance, not the runtime allowance. Width is the unrounded floating-point upper minus recovered secant value; a tiny negative width reflects numerical error and is not a rational zero-width proof. The full records retain both solver objectives, bounds, statuses, residuals, and allowances.